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Detection DNS Tunneling Botnets

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Abstract—**Botnets are often used in cyberattacks on network services and individual users, so the ability to detect botnets is very important. Botnets use DNS tunneling to send malicious command-and-control (C&C) commands to victims' hosts. Unfortunately, DNS tunneling attacks are very hard to detect. The paper presents a new approach for DNS tunneling botnet detection, which considers all the features and architectural characteristics of botnets. The technique described in this paper is highly efficient at detecting DNS tunneling attacks.**

Keywords—malware, botnet, botnet detection, DNS, DNS tunneling attacks, networks, classifier, network security

I. INTRODUCTION

The modern digital world is subject to a huge number of different cyberattacks. Botnets are one of the most powerful and common sources of cyberattacks today. Most botnets use the Domain Name System (DNS) to hide the command-and-control server (C&C server) and to provide reliable communication between the infected devices connected to the botnet and the C&C [1-3]. The DNS system is used to resolve the domain name of the C&C server, which can use various methods of evading detection based on the use of DNS [4-5].

The Domain Name System (DNS) is a protocol that was originally intended and developed exclusively for the resolution of domain names. Based on this, DNS traffic was usually not considered as a way to organize malicious communications and was not analyzed for their presence. This provided for the possibility of organizing a hidden bidirectional data transmission channel using the DNS protocol. This led attackers to implement such channels using DNS tunneling technology. This technology allows the botnet owner to encode malicious communications traffic within DNS requests without being noticed. The malicious traffic be either botnet command and control traffic or stolen confidential information from an infected device. To be successful, the attacker must control the domain and the authoritative DNS server for this domain. This makes it possible to manage tunneling on the side of the DNS server, control DNS requests from an infected device, decode encrypted data received in DNS requests, and generate encoded commands intended for an infected device only using DNS responses.

An important advantage of DNS tunneling technology for an attacker is the fact that no direct connection is established between the infected device and the attacker's DNS server. Also, the transmission of tunneled data between the infected device and the attacker's server occurs through legitimate DNS servers. In this situation it is important to deeply discover the principles the DNS tunneling and to develop the approach for detection of botnets, that use the DNS tunneling taking into account features and architectural characteristics of botnets.

II. RELATED WORKS

There are many different approaches to botnet detection [6] based on machine learning [7-8]. In [9-11] the life cycle of botnets is considered in terms of their harmful activities.

In [12], the taxonomy and life cycle (LC) of botnets are presented using a product development model. According to this model, the LC of the botnet consists of stages: Concept, Recruitment, Interaction, Marketing and Execution of the attack, giving rise to the acronym CRIME)

One of the earliest models of malware is a simplified mathematical epidemiological model [13]. An improvement on this model is the Susceptible-Infected-Control (SIC) model, which describes botnets using a stochastic model of population evolution. In [10], the SIC detection model assumption is that all possible hosts are susceptible to infection by botnet malware, and there is little chance that the device will be infected and become a controlled botnet. The model provides two goals - to track the dynamics of population growth of the botnet and to compare the results of different strategies to mitigate botnet attacks.

In [14] an approach "Susceptible-Infected-Susceptible" (SIS) was proposed. The developed model applies the concept of vaccination rate. Known potential vulnerabilities in computer systems or vulnerabilities identified through reverse engineering of malware samples may be vectors of infection. After determining the vector of infection as a countermeasure, a special

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exploit can be created for it, which eliminates the vulnerability - "vaccination".

In [15], deep learning methods are used to model packet flows in P2P botnets. Deep learning provides more accurate detection of traffic structure than non-hierarchical machine learning.

In [16], stochastic simulation was used to simulate a DDoS attack on the Cloud Data Center (CDC). This model uses modules to filter traffic within bandwidth thresholds and load balancing, and queue data center optimization to reduce request flows.

In [17] authors use a detection model based on evolutionary games and an epidemiological model to analyze the resilience of interacting botnets in situations where botnet owners interact in collaboration or competition.

In [18] a dynamic game model is proposed, in which the attacker is a bot-master (owner of the bot-network), who initiates a DDoS-attack, and the defender is a firewall. The model assumes that players can act both rationally and irrationally. The purpose of this model is to improve the performance of firewalls. [19] uses the theory of middle field games in combination with stochastic processes and epidemiological states of infection to model the spread of botnets depending on the decisions of the owner of the device regarding the protection system.

In [20], a nonparametric Bayesian model was used to separate bot traffic from harmless traffic. The difficulty of identifying malicious traffic is that the communication schemes of botnets are constantly evolving, and malicious traffic is transmitted simultaneously with harmless traffic. The proposed model associates bytes associated with traffic and communications, with specific botnets or with good quality programs. To represent the activity of the botnet, a hidden Markov model is used, which tries to detect harmful traffic to mitigate the attack of the botnet.

In [21], the interaction of social network users is modeled in the form of a graph, which is used to visualize traffic, model the behavior of programs, and detect anomalies or attacks of botnets.

Economic models [22-24] try to mitigate the threat of botnets, reducing the ability of botnet owners to make a profit.

Usually, botnet detection is only possible during or after a botnet attack [25-27]. Well-known approaches provide an opportunity to test existing assumptions and mitigate the effects of botnet attacks. Given the complexity of identifying malicious bot traffic, as well as the fact that most botmasters use DNS services to manage botnets and evade detection of botnets, the urgent task is to develop a model of botnets based on DNS tunneling. This allows botnets to be detected before the attack. Our approach considers of the features and architectural characteristics of a botnet.

III. Approach for DNS Tunneling Botnet Detection

A. DNS-based botnet lifecycle

To develop our DNS tunneling botnet detection technique we have to start with a botnet life cycle model. Botnets must be able to:

(1) infect the target host;
(2) register itself in the infected system and establish the connection with the other bots in the botnet and its C&C;
(3) execute built-in commands of the infected system;
(4) perform updates;
(5) delete itself after its mission is over.

Every bot in a botnet can infect the victim hosts. In the infection phase, the existing protection mechanisms of the computer system are overcome, a malicious script (shellcode) is executed, and a bot program is installed on the infected computer system. Infection of the computer system with a bot can be a consequence of a successful attack or happen later in a botnet's lifecycle. One of the most common ways that a computer gets infected is through a drive-by download which occurs when a computer system is infected while visiting a compromised website that contains malicious code. Malicious code can be downloaded to the hard disk of the computer system or embedded in dynamic libraries in the memory of a legitimate process without saving to disk. In the second case, the harmful process exists only in the operating memory of the infected computer system until the system is rebooted. Infection using a popular resource which is visited by the user regularly allows multiple infections of the computer system that will keep the computer in the botnet.

After a successful host infection, the botnet is ready to execute DNS requests to inform its C&C server about the successful infection. To do this bot tries to send recursive DNS request C&C server's domain name via the stub resolver. If the DNS request is successful, the bot is registered in the botnet and can start to communicate with its C&C server.

After the TTL DNS period is over or the infected host is rebooted, the bot starts to send DNS requests and tries to update the records of the local cache and the cache of the recursive DNS server.

Fig. 1 shows a communication scheme between botnet C&C server using DNS, where the sequence of requests and responses is noted from 1 to 10.

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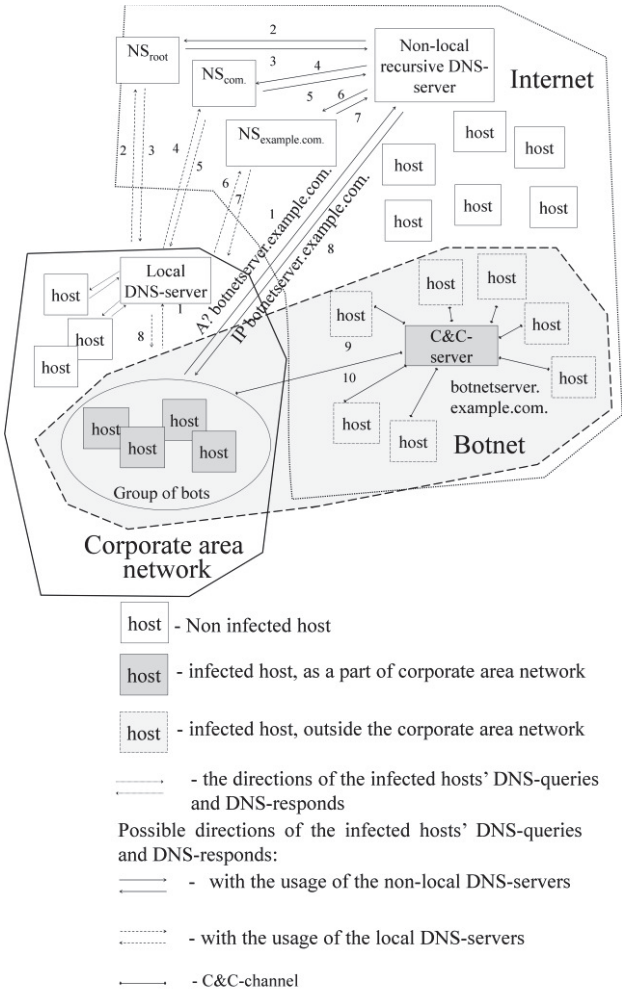


Figure 1. The communication scheme between bots and their C&C server using DNS

DNS-requests are started by the bot in the accompanying circumstances:

1) To establish connections with the other bots of a botnet after the successful infection of the host;

2) when bots need to send DNS queries after an unsuccessful connection attempt with the C&C server;

3) while the botnet is in the migration stage;

4) after the IP address of the C&C server was changed;

5) while engaging in malicious activities (DDoS-assaults, spam-mailings);

6) after the infected host is rebooted;

7) while trying to obtain additional domain names for its sustainability.

In the situation, when the domain name's DNS records were cached by the DNS client of the network host, the repeated DNS request don't go past the local DNS cache of the PC framework before the TTL springs up.

Many types of botnets overlook the TTL time frame, the span of which was contained in a reaction from the definitive DNS worker to the DNS demand. This implies that the bot clears the neighborhood DNS reserve and plays out a rehashed DNS demand for the area name until the finish of the TTL time frame, which permits expanding the adaptability and unwavering quality of bot network the board.

Fig. 2 demonstrates carrying out a repeated DNS-request for the domain name before the end of the TTL-period.

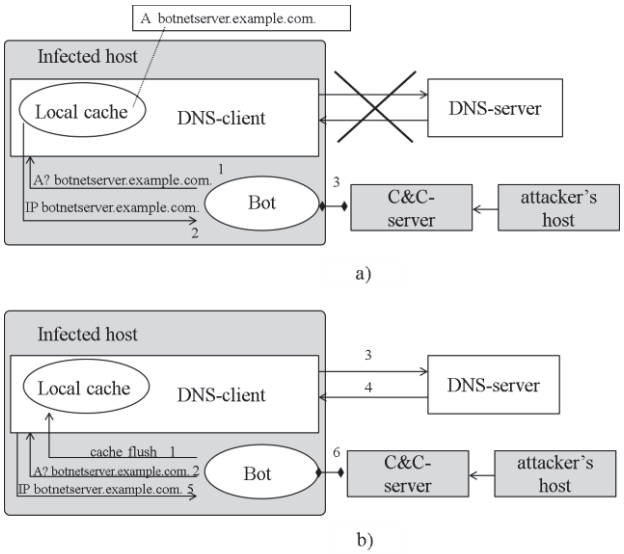


Figure 2. Carrying out repeated DNS-requests for the domain name before the end of the TTL-period: a) if there is a cached DNS-response for the domain name in the local DNS cache; b) after clearing the local cache DNS

B. DNS Tunneling Botnet Attack Model

When using DNS-tunneling technology an attacker uses a fake DNS server, which is an authoritative DNS name server for the DNS zone whose domain names are being queried by bots. This allows bots to send encrypted messages to the attacker's server in the form of DNS-queries regarding domain names of this zone, and bots receive commands from the attacker, also encoded as DNS-responses.

Fig. 3 demonstrates the transmission of malicious traffic via the fields of the DNS message, where the response is represented by 1… 4.

In this case set of domain names D DNS zone acts as an analogue of the domain names of the C&C server of the botnet, IP address e a fake DNS server that usually remains constant, $\{d_1,...,d_n\} \to e$.

The header of the DNS message can be described as follows:

$$\Lambda_{HD} = \left\langle \Lambda_{ID}, \Lambda_{OPC}, \Lambda_{RC}, \Lambda_{QDC}, \Lambda_{ANC}, \Lambda_{NSC}, \Lambda_{ARC} \right\rangle, \quad (1)$$

where Λ_{ID} - special ID number applied to perform the

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linking between the DNS-query and DNS-answer (ID field);

$\Lambda_{OPC} \in \{0,..,2\}$ - the type of DNS request (OPCODE field), 0 – standard type, 1 – inverse type, 2 - request concerning the server status;

$\Lambda_{RC} \in \{0,..,5\}$ - number that identify the code of response (RCODE field), 0 - response with no error, 1 - error that indicates the incorrect format, 2 – indication of the server failure, 3 - indications that the domain name does not exist, etc.;

Λ_{QDC} - value that indicates the number of records in the section of the request (QDCOUNT field);

$\Lambda_{ANC}, \Lambda_{NSC}, \Lambda_{ARC}$ - value that indicates the number of records concerning resources, servers' names, and additional information (ANCOUNT field, NSCOUNT field, ARCOUNT field).

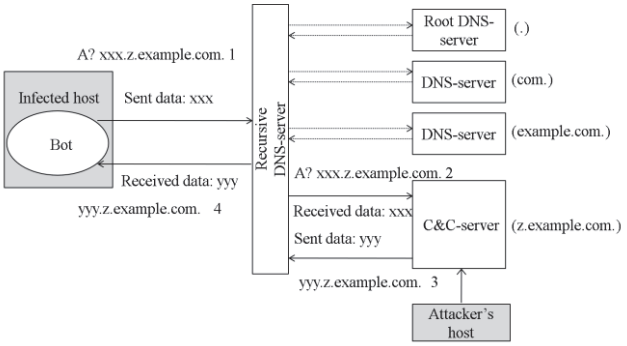


Figure 3. The scheme of transmission of the malicious traffic via the fields of the DNS messag

The sections of feedback, name servers and additional information have the same format and can be described as sets of resource records as follows:

$$\Lambda_S = \left\{ \left(\Lambda_{S,NM}, \Lambda_{S,TP}, \Lambda_{S,TTL}, \Lambda_{S,RDL}, \Lambda_{S,RDT} \right)_n \right\}_{n=1}^{N_{RR,S}}, \quad (2)$$

where $S \in \{"ANS","ATH","ADD"\}$, $\Lambda_{S,NM}$ - the name of the domain to which the resource record belongs (field NAME);

$\Lambda_{S,TP}$ - obtained value of the type of resource record code (TYPE field), which determines the value and data format in the response RDATA field;

$\Lambda_{S,TTL}$ - the value of the of DNS-records time to life (TTL field);

$\Lambda_{S,RDL}$ - the value that indicates the length of the RDATA field (RDLENGTH field);

$\Lambda_{S,RDT}$ - an information concerning the resource (RDATA field);

$N_{RR,S}$ - the value that indicates the number of resource records in the section of DNS-response (equal to the value $\Lambda_{ANC}, \Lambda_{NSC}, \Lambda_{ARC}$ for the relevant section).

Let us define the set of features concerning the DNS-response that can indicate the DNS tunneling attack as follows:

$$F = \left\{ f_1, f_2, f_3, f_4, f_5, f_6, f_{E_B} \right\}, \quad (3)$$

where f_1 - the value that indicates the length of the domain name;

f_2 - the value that indicates the number of different characters in the requested domain name;

f_3 - the value of the evaluated entropy concerning the domain name;

f_4 - the possible maximum obtained entropy value concerning the resource records of the observed DNS response;

f_5 - binary indication of the rarely used types of DNS records, or those that are not normally used by users;

f_6 - the value of the average size concerning obtained the domain name's DNS-messages;

f_{E_B} - the defined dependency function of the message field length on the DNS-message field's entropy, taking into account the specified n-coding.

Availability the application of DNS-based bot evasion detection technologies can be represented using knowledge that can begiven in the form of the following rules:

$$\begin{aligned} &if \left(\left(f_1 \in [75,255] \ and \ f_2 \in (27,37] \right) or \left(f_3 \geq f_{E_{B32}} \ or \right. \right. \\ &or \left. \left(f_4 \geq f_{E_{B64}} \ or \ f_4 \geq f_{E_{B256}} \right) or \ f_5 = 1 \right) \Big) and \\ &and \ f_6 > 300 \Rightarrow DNS_tunneling \end{aligned} \quad (4)$$

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IV. EXPERIMENTS

A. Evaluation and Experimental Setup

A set of experiments were performed to test out approach. Initially, a normal DNS dataset was created. After the dataset was created, DNS traffic within our test botnet consisting of more than 50 hosts was captured over a 6-month period. We used the dns2tcp DNS tunneling tool to perform DNS tunneling attacks.

To generate malicious traffic Iodine, DNScat2, and the dns2tcp DNS tunneling tools were employed [28-30].

The main purpose of the experiments was to determine the effectiveness of DNS tunneling botnet detection via distinguishing between normal and malicious DNS traffic. One of important aspects of the DNS traffic captures was the use of the DNS proxy to improve our ability to observe DNS requests to a local cache. To execute a DNS tunneling attack, we registered a server, and a domain name was employed.

The detection of DNS tunneling was done by the BotGrabber. BotGrabber can perform passive monitoring and active DNS probing of the network [31-35].

B. The Dataset

BotGrabber makes its decisions using the following set of classifiers: c-means, Support Vector Machine, Naïve Bayes, K-Nearest Neighbors, Random Forest [34, 36-39].

Our model of botnet DNS tunneling attacks made it possible to obtain efficient detection. Our results were obtained using the set of normal DNS requests and the set of malicious DNS requests. The malicious requests were obtained from the database in [40] and included those the generated via the previously mentioned DNS tunneling tools [28-30].

After DNS tunneling attacks by botnets were performed the cached traffic was analyzed by the BotGrabber system. During the experiments 12426 DNS requests were cached and processed.

C. Results

To evaluate the efficiency of the approach the following quantity measures were used:

1) True Positives (TP);
2) True Negatives (TN);
3) False Positives (FP);
4) False Negatives (FN);
5) sensitivity;
6) specificity;
7) overall accuracy, Q.

Mentioned measures were obtained as follow:

$$Sensitivity = TP/(TP+FN), \quad (5)$$

$$Specificity = TN/(TN+FP), \quad (6)$$

$$Q = (TP+TN)/(TP+TN+FP+FN). \quad (7)$$

The results demonstrate that all applied in the BotGrabber system classifiers were able to detect botnet DNS tunneling attacks with an accuracy of over 99% and with a false positive rate of less than 4%. Table 1 presents the results of our experiments.

TABLE I. THE DETECTION RESULTS

Classifier	TP	FN	TN	FP	Sensitivity, %	Specificity, %	Overall accuracy, %
c-means	17	1	12405	3	94,44	99,98	99,97
SVM	22	1	12399	4	95,65	99,97	99,96
Naïve Bayes	19	1	12403	3	95,00	99,98	99,97
K-Nearest Neighbors	16	2	12405	3	88,89	99,98	99,96
Random Forest	23	3	12399	1	88,46	99,99	99,97
Rotation Forest	20	1	12402	3	95,24	99,98	99,97

V. CONCLUSIONS

This paper presents a new approach for detecting DNS tunneling botnet attacks. In this paper the DNS-based botnet lifecycle was investigated and the communication scheme between botnet C&C server using DNS was explained. The proposed technique is based on our DNS tunneling botnet attack model which considers all botnet features and architectural characteristics.

To evaluate the effectiveness of the proposed technique a set of experiments were held that used DNS tunneling tools to generate malicious traffic.

BotGrabber was used to perform passive monitoring and active DNS probing of the network.

The results demonstrate that our technique was able to detect botnet DNS tunneling attacks with an accuracy of 99% and with false positives not exceeding 4%.

In order to improve the detection efficiency, the future work will be devoted to the development of new approaches that will involve other machine learning classifiers.

REFERENCES

[1] J. Shi, Y.-B. Leau, K. Li, J.H. Obit, "A comprehensive review on hybrid network traffic prediction model," *International Journal of Electrical and Computer Engineering*, vol. 11, issue 2, pp. 1450–1459, 2021.

[2] O. Olowoyo, P. Owolawi, "Malware classification using deep learning technique," *Proceedings of the 2nd International Multidisciplinary Information Technology and Engineering Conference IMITEC'2020*, 2020, 9334071.

[3] H. Qin, J. Yang, X. Li, Z. Luo, Q. Guo, "Research on DNS anomaly detection technology based on multiple features," *Journal of Shenzhen University Science and Engineering*, vol. 37, pp. 36–43, 2020.

[4] M. Abdelhaq, R. Alsaqour, M. Alaskar, M. Sehaibani, D. Moyna, "The resistance of routing protocols against DDOS attack in

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MANET," *International Journal of Electrical and Computer Engineering*, vol. 10, issue 5, pp. 4844–4852, 2020.
[5] X. Guo, Z. Pan, Y. Chen, "Application of passive DNS in cyber security," *Proceedings of the 2020 IEEE International Conference on Power, Intelligent Computing and Systems, ICPICS'2020*, 2020, pp. 257–259, 9202344.
[6] P. Wainwright, H. Kettani, "An analysis of botnet models," *Proceedings of the 2019 3rd International Conference on Compute and Data Analysis*, 2019. pp. 116-121.
[7] V. Efimov, I. Kotenko, & I. Saenko, "Network application-layer protocol classification based on fuzzy data and neural network processing," *International Journal of Computing*, vol. 19, issue 3, pp. 335–346, 2020. https://doi.org/10.47839/ijc.19.3.1877.
[8] A. Branitskiy, I. Kotenko and I. B. Saenko, "Applying machine learning and parallel data processing for attack detection in IoT," *IEEE Transactions on Emerging Topics in Computing*, doi: 10.1109/TETC.2020.3006351.
[9] M. Feily, A. Shahrestani, S. Ramadass, "A survey of botnet and botnet detection," *Proceedings of the 2009 Third IEEE International Conference on Emerging Security Information, Systems and Technologies*, 2009, pp. 268-273.
[10] M. Khosroshahy, M.K.M. Ali, D. Qiu, "The SIC botnet lifecycle model: A step beyond traditional epidemiological models," *Computer Networks*, vol. 57, no. 2, pp. 404-421, 2013.
[11] J. Liu, Y. Xiao, K. Ghaboosi, H. Deng, J. Zhang, "Botnet: Classification, attacks, detection, tracing, and preventive measures," *EURASIP Journal on Wireless Communications and Networking*, vol. 1, pp. 1-11, 2009.
[12] R. Rodríguez-Gómez, G. Maciá-Fernández, P. García-Teodoro, "Survey and taxonomy of botnet research through life-cycle," *ACM Computing Surveys (CSUR)*, vol. 201345, no. 4, pp. 1-33, 2013.
[13] J. Kephart, S. White, "Measuring and modeling computer virus prevalence," *Proceedings of the 1993 IEEE Computer Security Symposium on Research in Security and Privacy*, 1993. pp. 2-15.
[14] J. Jerkins, J. Stupiansky, "Mitigating IoT insecurity with inoculation epidemics," *Proceedings of the ACMSE 2018 Conference (ACMSE'18)*, 2018, vol. 4, pp. 1-6.
[15] J. van Roosmalen, H. Vranken, M. van Eekelen, "Applying deep learning on packet flows for botnet detection," *Proceedings of the 33rd Annual ACM Symposium on Applied Computing (SAC'18)*, 2018, pp. 1629-1636.
[16] I. El Mir, D. Kim, A. Haqiq, "Towards a stochastic model for integrated detection and filtering of DoS attacks in cloud environments," *Proceedings of the 2nd International Conference on Big Data, Cloud and Applications*, 2017, vol. 10, pp. 1-6.
[17] L. Song, Z. Jin, G. Sun, "Modeling and analyzing of botnet interactions," *Physica A: Statistical Mechanics and its Applications*, vol. 390, no. 2, pp. 347-358, 2011.
[18] Y. Wang, J. Ma, L. Zhang, W. Ji, D. Lu, X. Hei, "Dynamic game model of botnet DDoS attack and defense," *Security and Communication Networks*, vol. 9, no. 16. pp. 3127-3140, 2016.
[19] V. N. Kolokoltsov, A. Bensoussan, "Mean-field-game model for botnet defense in cyber-security," *Applied Mathematics & Optimization*, vol. 74, no. 3, pp. 669-692, 2016.
[20] J. Divita, R. Hallman, "An approach to botnet malware detection using nonparametric Bayesian methods," *Proceedings of the 12th International Conference on Availability, Reliability and Security (ARES'17)*, New York: Association for Computing Machinery (ACM), vol. 75, pp. 1-9, 2017.
[21] M. Faloutsos, "Detecting malware with graph-based methods: traffic classification, botnets, and Facebook scams," *Proceedings of the 22nd International Conference on World Wide Web (WWW'13 Companion)*. New York: Association for Computing Machinery (ACM), 2013, pp. 495-496.
[22] Z. Q. Li, A. Liao, "Striegel Botnet economics: Uncertainty matters," *Managing Information Risk and the Economics of Security*, Johnson, M. E. (Ed.). Springer, Boston, MA, 2009, pp. 245-267.
[23] G. Bottazzi, G. Me, "The botnet revenue model," *Proceedings of the 7th International Conference on Security of Information and Networks (SIN'14)*, New York: Association for Computing Machinery (ACM), 2014, pp. 459-465.
[24] Z. Li, Q. Liao, "Toward a monopoly botnet market," *Information Security Journal: A Global Perspective*, vol. 23, no. 4-6, pp. 159-171, 2014.
[25] M. Torquato, M. Vieira, "Moving target defense in cloud computing: A systematic mapping study," *Computers and Security,* vol. 92, 101742, 2020.
[26] S.H. Mousavi, M. Khansari, R. Rahmani, "A fully scalable big data framework for Botnet detection based on network traffic analysis," *Information Sciences*, vol. 512, pp. 629–640, 2020.
[27] T. Lange, H. Kettani, "On security threats of botnets to cyber systems," *Proceedings of the 2019 6th International Conference on Signal Processing and Integrated Networks, SPIN'2019*, 2019, pp. 176–183, 8711780.
[28] Iodine. Tool for tunneling IPv4 data through a DNS server. https://code.kryo.se/iodine.
[29] DNScat2. DNS Tunneling tool. https://github.com/iagox86/dnscat2.
[30] dns2tcp. Tool for relaying TCP connections over DNS. https://github.com/alex-sector/dns2tcp.
[31] S. Lysenko, O. Savenko, K. Bobrovnikova, "DDoS Botnet detection technique based on the use of the semi-supervised fuzzy c-means clustering," CEUR-WS, ISSN: 1613–0073 (Scopus), 2018, vol. 2104, pp. 688-695.
[32] S. Lysenko, K. Bobrovnikova, R. Shchuka, O. Savenko, "A cyberattacks detection technique based on evolutionary algorithms," *Proceedings of the 2020 IEEE 11th International Conference on Dependable Systems, Services and Technologies*, May 2020, pp. 127-132.
[33] S. Lysenko, K. Bobrovnikova, O. Savenko, R. Shchuka, "Technique for cyberattacks detection based on DNS traffic analysis," CEUR-WS, vol 2732, ISSN: 1613–0073, 2020, pp. 171-182.
[34] S. Lysenko, K. Bobrovnikova, O. Savenko, A. Kryshchuk, "BotGRABBER: SVM-Based self-adaptive system for the network resilience against the botnets' cyberattacks," In: Gaj P., Sawicki M., Kwiecień A. (eds) Computer Networks, CN 2019. Communications in Computer and Information Science, Springer, Cham, vol. 1039, 2019, pp. 127-143.
[35] O. Drozd, K. Zashcholkin, O. Martynyuk, O. Ivanova, J. Drozd, "Development of checkability in FPGA components of safety-related systems," CEUR Workshop Proceedings, vol. 2762, 2020, pp. 30-42. Online http://ceur-ws.org/Vol-2762/paper1.pdf.
[36] Y.P. Kondratenko, & D. Simon, "Structural and parametric optimization of fuzzy control and decision making systems," *In Recent Developments and the New Direction in Soft-Computing Foundations and Applications*, Springer, Cham, 2018, pp. 273-289.
[37] A. Cabri, G. Suchacka, S. Rovetta and F. Masulli, "Online Web Bot Detection Using a Sequential Classification Approach," 2018.
[38] S. Ustebay, Z. Turgut and M.A. Aydin, "Intrusion detection system with recursive feature elimination by using random forest and deep learning classifier, *Proceedings of the 2018 International Congress on Big Data, Deep Learning and Fighting Cyber Terrorism (IBIGDELFT)*, Ankara, Turkey, 2018, pp. 71-76.
[39] T. Sochor, & M. Zuzcak, "Attractiveness study of honeypots and honeynets in internet threat detection," *Communications in Computer and Information Science*, Springer, Cham, 2015. pp 69-81.
[40] Canadian Institute for Cybersecurity. Botnet dataset, https://www.unb.ca/cic/datasets/botnet.html (acces.January, 2021).

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